Biochemistry

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Biochemistry (A06222)

Short Title: Biochemistry
Faculty Science and Computing
Department Science
Cluster Biology
Author KTAMBLING
Credits 5 **Level:** Introductory

Description of Module / Aims

This module builds on students' knowledge of basic biochemical concepts obtained in previous modules (Cell Biology and Biochemistry; Organic Chemistry and Biomolecules), focusing particularly on proteins, enzymes and metabolic reactions. The module covers protein synthesis in cells by the process of translation, enzyme properties and enzyme kinetics, as well as energy metabolism, and an overview of pH and buffers. The theory is reinforced with practical content, including preparation of solutions, biochemical analyses and quantification or purification of proteins and enzymes.

Programmes

`BIOL-0011`	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (`WD_SMBIO_B`)	2 / 4 / M
`BIOL-0011`	BSc in Molecular Biology with Biopharmaceutical Science (`WD_SMBIO_D`)	2 / 4 / M

Indicative Content

- Acids, bases, pH and buffers: calculations and practical methods involved in the preparation of buffers
- General properties and classification of enzymes: introduction to enzyme kinetics and enzyme inhibition; assessment of enzyme activity and purity
- Translation: the genetic code and its role in protein synthesis
- The mechanism, role and control of the major metabolic pathways of the cell
- Practical element reflecting the indicative content of the module including: preparation of buffers and other solutions; quantitative biochemical assays; enzyme assays and examining enzyme kinetics and activity

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Explain key concepts of pH and buffers.
2. Relate enzyme structure and classification to catalytic activity.
3. Describe the basics of enzyme kinetics and identify the effects of different classes of reversible inhibitors on enzyme kinetics.
4. Recognise the principles and mechanisms involved in the translation of proteins.
5. Describe the principle pathways of energy metabolism and outline how these pathways may be regulated.
6. Complete experiments in biochemical and enzyme assays and interpret the data obtained.

Learning and Teaching Methods

- Lectures.
- Library work (self-directed study).

- Practical laboratory work.
- Student engagement will be enhanced through a blended teaching approach, combining on-line learning through interactive use of the Moodle virtual e-learning environment with lectures and videos/animations.
- Feedback will be provided on an on-going basis to encourage reflective, critical and independent learning.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,4,5
Continuous Assessment	50%	
Practical	50%	6

Assessment Criteria

<40%: No understanding of the basics of biochemistry. Unable to carry out critical thinking or practical skills.

40%-49%: Is able to understand basic biochemistry. May have some difficulty with critical thinking and problem solving.

50%-59%: All of the above including a good understanding of theoretical biochemistry and demonstration of good practical skills.

60%-69%: Show a high level of competence and efficiency in theoretical and practical biochemistry.

70%-100%: All of the above to an excellent level. Has the ability to describe and discuss theoretical and practical biochemistry.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	36	
Independent Learning	75	

Essential Material

- "Moodle used to provide links to useful websites." https://moodle.wit.ie
- Campbell , N.A., J.B. Reece, L.A. Urry, M.L. Cain and S.A. Wasserman. _Biology_. 8th ed. San Francisco: Pearson, Benjamin Cummings, 2008.
- Voet, D., J.G. Voet and C. Pratt. _Fundamentals of Biochemistry_. 2nd ed. Hoboken, N.J.: Wiley, 2006.

Supplementary Material

- "Biochemistry Free and Easy Resources." http://biochem.science.oregonstate.edu/content/biochemistry-free-and-easy

- Mathews, C.K., K.E. van Holde, D.R. Appling and S.J. Anthony-Cahill. _Biochemistry_. 4th ed. USA: Prentice Hall, 2013.
- Nelson, D.L. and M.M. Cox. _Lehninger__Principles of Biochemistry_. 6th ed. New York: W.H. Freeman, 2012.
- Palmer, T. _Enzymes: Biochemistry, Biotechnology and Clinical Chemistry (Horwood Chemical Science Series)_. 5th ed. USA: Horwood Publishing Limited, 2001.

Requested Resources

- Lecture Room: Loose Seated
- Room Type: Biology Lab